RadarGPT: Contact-free Cardiac Diagnosis Based on mmWave Radar and LLM

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***Abstract*—Continuous cardiovascular health monitoring is crucial for the early detection of heart diseases, but traditional contact-based method faces daily deployment challenges due to user discomfort. Recently, millimeter-wave (mmWave) radar has emerged as a promising contact-free alternative. However, existing radar-based methods primarily focus on extracting simple physiological metrics or reconstructing waveforms, lacking the ability to directly provide intuitive and interpretable natural-language diagnostic reports. In this paper, we propose RadarGPT, a contact-free cardiac diagnosis system based on mmWave radar and Large Language Model (LLM). To isolate cardiac-related radar signals, we design a robust signal processing pipeline. To address the profound modality and domain knowledge gap despite data scarcity, we introduce a two-stage training strategy with a physiological prior-guided data augmentation method. Specifically, a cross-modal contrastive pretraining stage aligns radar signal representations with ECG and textual semantics, followed by a LoRA-based fine-tuning stage to empower the LLM for direct diagnostic reasoning. We implement RadarGPT on a commercial mmWave radar, and the results demonstrate that it achieves high standards in both physiological metrics estimation and coherent clinical report generation, demonstrating promising generalization to unseen subjects.**

***Index Terms*—Wireless sensing, mmWave radar, cardiac diagnosis, large language model, deep learning.**

I. Introduction

Continuous, long-term cardiovascular health monitoring is of great importance for the early detection of heart diseases. While the Electrocardiogram (ECG) remains the clinical gold standard [1], its reliance on skin-contact electrodes inevitably causes discomfort during prolonged use, making it unsuitable for daily use and individuals with skin sensitivity. Recently, millimeter-wave (mmWave) radar has emerged as a promising alternative for contact-free vital sign monitoring [2]. However, existing radar-based methods primarily focus on extracting simple physiological metrics, such as average heart rate (HR) [3] and heart rate variability (HRV) [4], or utilize data-driven approaches to reconstruct ECG waveforms [1]. Although they can capture vital signs or recover fine-grained ECG waveforms, they require the cardiologist for interpretation and lack the ability to directly provide interpretable diagnostic reports.

The recent evolution of Large Language Models (LLMs), such as Qwen [5], offers a compelling solution to generate such reports. Unlike conventional task-specific models, LLMs provide a flexible interface for integrating sensing information and generating human-readable reports. Several works have fine-tuned LLMs to directly analyze ECG signals and output diagnostic results [6], [7], demonstrating that LLMs exhibit remarkable cross-modal reasoning capabilities. Intuitively, a straightforward extension for contact-free cardiac diagnosis would be to reconstruct ECG waveforms from radar echoes and subsequently feed them into an ECG-fine-tuned LLM. However, this cascading approach is highly complex and inefficient. It requires rigorous training to guarantee high-fidelity intermediate ECG reconstruction, followed by LLM fine-tuning to recognize the generated ECG waveforms reliably. This inspires us to ask a critical question: How can we empower LLMs to understand radar signals and directly generate clinically interpretable diagnostic reports?

To answer this question, we must address two fundamental challenges inherent in this direct radar-to-report paradigm. The first one is the significant modality gap among them. The mmWave radar captures the mechanical vibrations of the chest wall, which are often heavily obscured by respiration and body micro-motions. In contrast, ECG records the electrical activity of the heart, while diagnostic reports reside purely in the textual domain. Aligning the radar signals with the semantic space of ECG and text is essential for cross-modal comprehension. Another one is the substantial domain knowledge gap. While LLMs excel in natural language processing, they lack inherent exposure to radio frequency (RF) sensing data. Consequently, empowering LLMs to interpret these complex signals relies heavily on extensive domain-specific fine-tuning. This critical process, however, is severely hindered by the scarcity of synchronized large-scale datasets comprising radar, ECG and reports, particularly for unpredictable abnormal cardiac events.

In this paper, we propose RadarGPT, a contact-free cardiac diagnosis system using mmWave radar. To isolate the cardiac signals from human motions and respiration, we design a comprehensive signal processing pipeline combining B-spline interpolation and temporal differentiation techniques. To bridge the modality gap among radar, ECG and report representations, we propose a cross-modal alignment pretraining strategy that aligns them in a shared latent space. To ensure the LLM comprehends these radar semantics despite data scarcity, we introduce a physiological prior-guided data augmentation method and fine-tune the LLM using Low-Rank Adaptation (LoRA), enabling it to accurately translate radar signal features into reliable clinical reports. The main contributions are summarized

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as follows.

- To the best of our knowledge, RadarGPT is the first work to integrate mmWave radar sensing with an LLM for contact-free cardiac diagnosis. By leveraging comprehensive signal processing, cross-modal alignment and LoRA-based LLM fine-tuning, RadarGPT can directly comprehend radar signals and autonomously generate readable and intuitive cardiac diagnostic reports.
- To effectively isolate cardiac-related radar signals, we design a robust signal processing pipeline employing B-spline and temporal differentiation techniques. Building upon this, we propose a cross-modal pretraining strategy that aligns radar, ECG, and textual representations within a shared latent space, effectively bridging the inherent modality gap among them.
- To empower the LLM to comprehend the aligned radar features, we introduce a LoRA-based fine-tuning strategy. To support this fine-tuning process against the severe scarcity of abnormal cardiac data, we incorporate a physiological prior-guided data augmentation method. We implement RadarGPT using a commercial mmWave radar and the Qwen model. Results show that it achieves a high diagnostic accuracy of 96.05% across four distinct cardiac states, demonstrating promising subject-independent generalization capabilities on unseen subjects.

II. Preliminaries

A. Cardiac Motion Sensing based on FMCW Radar

To capture the subtle cardiac motion, we leverage the Frequency Modulated Continuous Wave (FMCW) radar. It continuously transmits sinusoidal signal with a linearly increasing frequency, known as a "chirp" signal, and receives the echo signals. The transmitted signal can be mathematically defined as

$$S_{TX}(t) = \exp\left[j(2\pi f_c t + \pi K t^2)\right], \tag{1}$$

where f_c denotes the initial frequency, and K represents the frequency slope of the chirp signal. Assuming the distance between radar and chest is $R(t) = R_0 + x(t)$, where R_0 represents the original distance and $x(t)$ denotes the small motion of the chest. The reflected signal $S_{RX}(t)$ is received by the antenna after a time delay of $\tau = \frac{2R(t)}{c}$. Then, a mixer combines the transmitted and received signals, generating the Intermediate Frequency (IF) signal as [8]

$$\begin{aligned} S_{IF}(t) &= S_{TX}(t) \cdot S_{RX}^*(t) \\ &\approx \alpha \exp\left[j2\pi\left(\frac{2f_c R(t)}{c} + \frac{2KR(t)}{c}\cdot t\right)\right], \end{aligned} \tag{2}$$

where α denotes the path loss.

By applying a fast Fourier transform to the IF signal [8], we can map the signal into the frequency domain as

$$\begin{aligned} Y(f) &= \mathcal{F}\{S_{IF}(t)\} \\ &= \alpha \exp\left[j\frac{4\pi f_c R(t)}{c}\right] \cdot \delta\left(f - \frac{2KR(t)}{c}\right) \\ &\overset{\text{def}}{=} y(t)\delta\left(f - \frac{2KR(t)}{c}\right), \end{aligned} \tag{3}$$

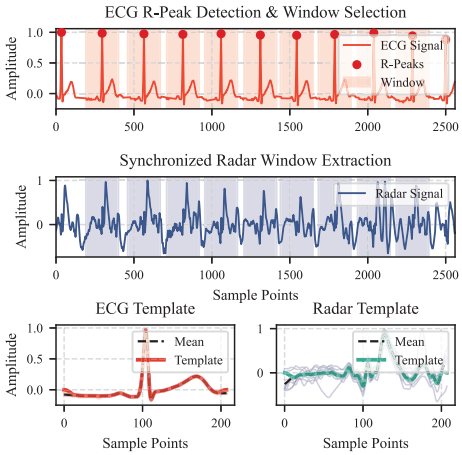


Fig. 1. Self-similarity visualization.

Here, $\delta(\cdot)$ represents the Dirac delta function. It indicates a linear relationship between the frequency f and reflection distance $R(t)$, which allows for the extraction of complex-value signal $y(t)$ corresponding to a specific distance. For the specific range corresponding to the chest, $y(t)$ yields a phase signal $\phi(t)$ that is highly sensitive to subtle motions

$$\begin{aligned} \phi(t) &= \frac{4\pi f_c R(t)}{c} \\ &= \frac{4\pi R_0}{\lambda} + \frac{4\pi x(t)}{\lambda} \overset{\text{def}}{=} \phi_0 + \Delta\phi(t), \end{aligned} \tag{4}$$

where $\lambda = \frac{c}{f_c}$ is the wavelength. The first term ϕ_0 is a constant phase shift determined by the static distance to the chest, while the second term $\Delta\phi(t)$ encapsulates the dynamic phase variation explicitly caused by the chest motion. Then, the chest motion $x(t)$ can be linearly reconstructed from the phase variation, which can be expressed as

$$x(t) = \frac{\lambda}{4\pi}\Delta\phi(t). \tag{5}$$

Since the chest motion contains the body, respiratory and cardiac motion, it is necessary to process the extracted radar phase signal to obtain useful cardiac motion related component, which will be specified in the following sections.

B. Physiological Priors Guided Data Augmentation

Due to the scarcity of large-scale, open-source datasets containing synchronized radar signals, ECGs and corresponding clinical reports, we have to construct our own dataset through real-world collection. Moreover, since abnormal cardiac events are naturally infrequent, this would inevitably yield heavily imbalanced datasets dominated by normal rhythms. To mitigate the inherent data scarcity and imbalance, we introduce a physiological prior guided data augmentation method. The core idea stems from the temporal self-similarity of cardiac motion [4]. Specifically, despite varying cycle lengths induced by different physiological conditions, the fundamental morphological structure of individual heartbeats remains relatively consistent for the same subject. Therefore, our augmentation method involves extracting a representative heartbeat template of radar and ECG signals as shown in Fig. 1 and resampling it to simulate physiologically realistic cycle length. To achieve it, we detect R-

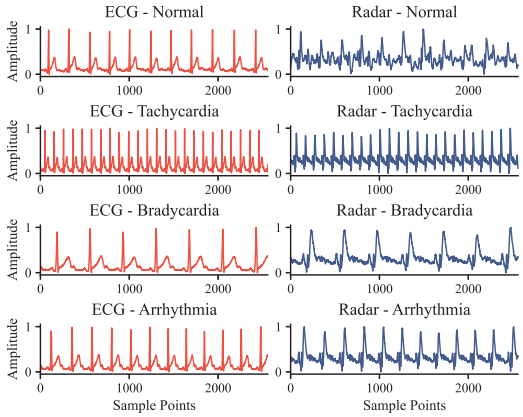


Fig. 2. ECG and radar signal examples based on data augmentation.

peaks in ECG signal to segment a continuous period of cardiac motion from the same subject into distinct cardiac cycles using a window based on the mean R-R intervals. These cycles are then aligned and averaged to generate a representative heartbeat template for both ECG and radar signals.

To enrich the data with diverse pathological patterns, we dynamically warp these templates via linear interpolation to match the desired cycle length [4]. We focus on four distinct cardiac conditions based on heart rate and the percentage of successive inter-beat intervals (IBI) that differ by more than 50 ms (pNN50) [4], which is defined as follows.

- **Normal.** It exhibits regular, moderate cycles ranging from 60 to 100 beats per minute (bpm), and the pNN50 $<$ 20.
- **Tachycardia.** It is characterized by shortened IBIs, with a heart rate exceeding 100 bpm, and the pNN50 $<$ 20.
- **Bradycardia.** It is marked by prolonged IBIs, with a heart rate dropping below 60 bpm, and the pNN50 $<$ 20.
- **Arrhythmia.** It presents irregular and unstable IBIs, where the pNN50 $>$ 20.

Additionally, random amplitude scaling factors and noises are applied to each warped heartbeat to introduce physiological variability. These scaled and warped individual heartbeats are then concatenated to form a continuous synthetic signal using the Tukey window [9]. In Fig. 2, we show an example of the ECG and radar phase signals based on our data augmentation method. These results demonstrate that it provides physiologically plausible rhythm variations for alleviating data scarcity. Although the augmentation strategy increases rhythm diversity, the generated abnormal patterns may not fully represent real pathological cases. Next, we will introduce the specific design of our system.

III. System Design

In this section, we present the details of RadarGPT, a contact-free cardiac diagnosis system using mmWave radar. The overall process is shown in Fig. 3. First, the mmWave radar captures the IF signals and an ECG monitor obtains the ECG signals simultaneously. Then, through signal preprocessing, we obtain the cardiac motion related radar phase signals. Based on the obtained radar-ECG pairs, we use physiological priors guided data augmentation method to synthesize abnormal radar-ECG samples and then obtain the corresponding clinical reports. Next, a two-stage training strategy is employed to develop a radar encoder and fine-tune a Large Language Model (LLM). This framework enables the system to interpret the radar phase signals and autonomously generate the clinical reports. After the training stages, RadarGPT is capable of performing online inference of a subject's cardiac status based solely on the input radar phase signals.

A. Data Collection and Signal Preprocessing

We perform data collection utilizing a commercial mmWave radar, which transmits FMCW signals and obtains the IF signals using the onboard mixer. The collected raw data are subsequently structured into a three-dimensional tensor organized by fast-time samples, chirps and antennas. Then, to determine the range and angle of the target people, we first perform fast fourier transform along the fast-time dimension, called Range-FFT [10], to the IF signal for each chirp, and then employ digital beamforming (DBF) techniques [11]. DBF is achieved by multiplying the steering vector to the received signals on different antennas after Range-FFT, resulting in beamformed signals. After that, we obtain the range-angle map (RAM) for each chirp. Then, to mitigate the static background reflections, we apply a Direct-current Component Filtering (DCF) method [10] to calculate the mean RAM across all chirps and subtract it from each RAM. Then, we employ the 2D Cell-Averaging Constant False Alarm Rate (2D CA-CFAR) detection algorithm [12] to determine the human target location by adaptively filtering out the noise points with low reflection power.

Based on the determined location of the human, we extract the raw phase signal and unwrap it using the Modified Differential and Cross-Multiply (MDACM) [13] technique to avoid phase discontinuity. Next, to suppress the effect caused by the body micro-motions, we apply the B-spline interpolation [14] to model and subtract this motion-induced phase distortion from the original phase signal. Furthermore, to remove the respiration introduced component in the phase signal, we employ temporal differentiation method, which enhances the cardiac-related signal compared to respiratory one. Finally, we obtain the processed cardiac-related phase signal, which is used to guide the LLM to generate the corresponding clinical report. Simultaneously, synchronized ECG signals are acquired to serve as the ground truth for signal analysis and automated report generation based on predefined templates. By augmenting the data based on the self-similarity as mentioned in Section II-B, we synthesize a large-scale radar-ECG-report dataset, comprising four cardiac rhythm categories. To adapt this dataset for LLM training, we organize the radar signals, ECG signals and their corresponding reports into a Question-Answering (QA) format. The input instructions use diverse templates to prompt for cardiac assessments as shown in Table I. The target answer is the generated clinical report, which contains detailing vital signs, diagnostic reasoning and clinical advice. Next, we will present the our training strategy based on this datasets.

B. RadarGPT Training Strategy

1) Radar Encoder Pretraining: While LLMs demonstrate remarkable capabilities in processing natural language and

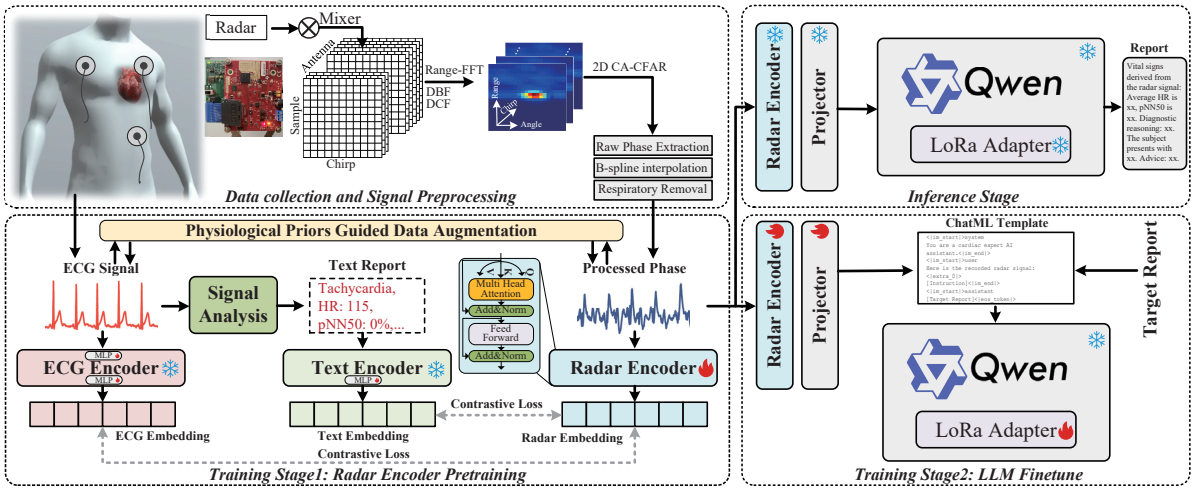


Fig. 3. Overall process of RadarGPT.

TABLE I
INSTRUCTIONS FOR HEART REPORT GENERATION.

ID	Instructions
1	Please generate a heart report based on the following radar data.
2	Analyze this radar signal and provide a cardiac health report.
3	I need a detailed heart rhythm report from this radar scan.
4	What does the radar data indicate about the subject's heart condition?
5	Provide a cardiac diagnostic report for the recorded radar signal.
6	Based on the radar monitoring, give me the heart status report.
7	Can you create a heart rate and rhythm report from this data?
8	Detect any heart abnormalities in this radar signal and write a report.
9	Output the heart health analysis report for this subject.
10	Please summarize the cardiac metrics and rhythm in a report.

conventional modalities, directly feeding raw mmWave radar phase signals into an LLM is likely to yield meaningless outputs and severe hallucinations. Inspired by successful cross-modal alignment methodologies [6], we utilize a contrastive learning framework to obtain a radar encoder to extract informative features from the radar phase signals. Rather than relying on a single reference modality, we project the radar phase signals into a shared semantic space alongside both ECG signals and clinical reports. Specifically, the radar encoder employs a 1D Transformer architecture, utilizing the self-attention mechanisms to extract global temporal representations [10]. To provide a stable reference space, we employ BioBERT [15], a pretrained language representation model in biomedical domain, alongside a pretrained ECG encoder [7] trained on multiple large-scale ECG datasets. The core parameters of these pretrained models are frozen, with additional lightweight projection layers are added and trained to accommodate the respective input and output dimensions. By optimizing the contrastive loss [6], the radar features are aligned with both the ECG and text features in the latent space. Specifically, for a given batch of size N, let r_i, e_i, and t_i denote the L_2-normalized embeddings of the radar signal, ECG signal, and text report for the i-th sample, respectively. The alignment between radar and ECG is optimized using a symmetric InfoNCE loss

$$\mathcal{L}_{r2e} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(r_i^T e_i/\tau)}{\sum_{j=1}^{N}\exp(r_i^T e_j/\tau)}, \tag{6}$$

$$\mathcal{L}_{e2r} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(e_i^T r_i/\tau)}{\sum_{j=1}^{N}\exp(e_i^T r_j/\tau)}, \tag{7}$$

where $\mathcal{L}_{r2e}$ and $\mathcal{L}_{e2r}$ are the contrastive losses for radar-to-ECG and ECG-to-radar, respectively, and τ is a learnable scaling parameter. The bidirectional radar-ECG contrastive loss is computed as

$$\mathcal{L}_{Radar-ECG} = \frac{1}{2}(\mathcal{L}_{r2e} + \mathcal{L}_{e2r}). \tag{8}$$

Similarly, by replacing the ECG embeddings e with the text embeddings t, we can obtain the symmetric InfoNCE losses $\mathcal{L}_{r2t}$ and $\mathcal{L}_{t2r}$ for radar-to-text and text-to-radar, respectively. The bidirectional contrastive loss between radar and text is formulated as

$$\mathcal{L}_{Radar-Text} = \frac{1}{2}(\mathcal{L}_{r2t} + \mathcal{L}_{t2r}). \tag{9}$$

The overall pretraining objective for the radar encoder is to simultaneously encode physiological morphology from the ECG and diagnostic semantics from the text, and the loss function can be expressed as

$$\mathcal{L}_{con} = \mathcal{L}_{Radar-ECG} + \mathcal{L}_{Radar-Text}. \tag{10}$$

2) LLM Finetune: After the radar encoder pretraining, we choose Qwen3-4B-Instruct-2507 [5] as our LLM backbone. Since the raw radar representations and the semantic text representations are in different feature spaces, we employ a two-layer Multilayer Perceptron (MLP) with a GELU activation function to project the radar features to match the hidden dimension of the LLM. The LLM is finetuned with Low-Rank Adaptation (LoRA) [16]. During fine-tuning, we unfreeze the entire radar encoder. This allows the radar phase signal representations to be dynamically refined through backpropagation from the LLM's task-specific losses. We apply LoRA adapter (rank=16 and α = 32) across query, key and value within all self-attention modules, as well as the projection layers in the feed-forward networks of the LLM.

To align with the instruction-following format of the base LLM, we structure our multimodal inputs using the standard ChatML template [5]. Specifically, the input sequence is

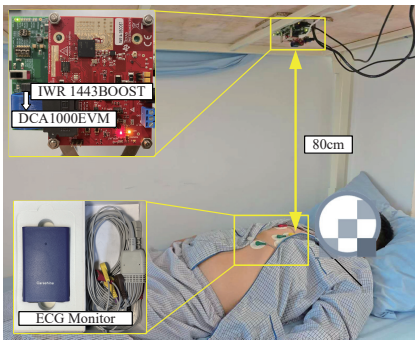


Fig. 4. Experimental scenario.

wrapped with specific control tokens (i.e., <|im_start|> and <|im_end|>) to explicitly define the roles of the system, user and assistant. A specific placeholder token, denoted as <|extra_0|>, is inserted into the user prompt and dynamically replaced by the continuous output embeddings from the radar MLP projector. Furthermore, an End-of-Sequence (EOS) token is appended directly after the target report to ensure the model learns to properly terminate its auto-regressive generation. During the fine-tuning process, we apply the standard causal language modeling objective [17], computing the loss based on the assistant's response.

IV. Experiments

A. Implementation Platform and Datasets

We evaluate the proposed RadarGPT system using the TI IWR1443BOOST mmWave radar [18], utilizing two transmitting antennas and four receiving ones to form an eight-channel virtual antenna array along the horizontal direction. As shown in Fig. 4, we collect synchronized radar-ECG data (250 Hz) from three subjects for 20-60 minutes. The radar configuration is summarized in Table II. After signal preprocessing, the processed radar phase and ECG signals are acquired and we use random cropping strategy with a 10.24-second sliding window to obtain 2,000 normal rhythm segments. Through data analysis and augmentation, we generate 2,000 segments for each of the other three cardiac categories, yielding a total of 8,000 radar-ECG-report samples for each participant. To evaluate the effectiveness of our system, we adopt a subject-independent evaluation protocol: data from two subjects are used for training, while the entirely unseen third subject is reserved for testing.

B. Evaluation Metrics

Since cardiac diagnostic reports are composed of specific clinical labels and vital signs [7], relying solely on traditional text similarity metrics may be insufficient. Thus, we evaluate our model across three dimensions: clinical classification [7], physiological parameter estimation and natural language generation (NLG) [6], respectively. Diagnostic classification performance is measured by overall accuracy (Acc), while Mean Absolute Error (MAE) is utilized to quantify the numerical deviation in predicting physiological metrics (Heart Rate and pNN50). To assess the quality of the generated textual reports, we employ BLEU-1 and BLEU-4 to measure word-level and phrase-level matching precision, ROUGE-1 and ROUGE-L for term recall and structural sequence overlap, and METEOR to evaluate broader semantic equivalence.

TABLE II
Radar Configuration.

Parameter	Value	Parameter	Value
Starting frequency	77.4 GHz	Maximum range	5.4 m
Range resolution	4.22 cm	Chirp duration	53.3 μs
ADC sampling rate	2400 ksps	ADC samples / chirp	128
Slow-time sampling rate	250 Hz	Segment duration	10.24 s

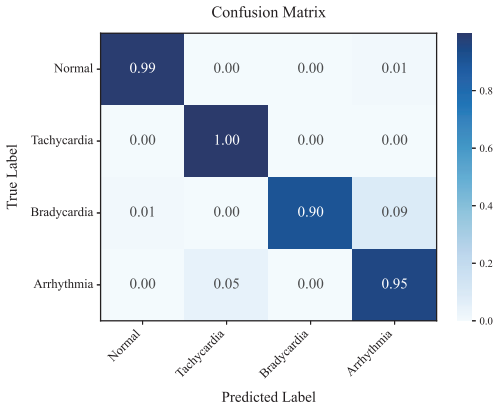


Fig. 5. Confusion Matrix.

C. Performance Evaluation

In Fig. 5, we show the confusion matrix for our model's clinical classification, where the diagnostic predictions are extracted from the generated report using keyword matching and regular expressions. The RadarGPT achieves high accuracy across all four cardiac rhythm categories, demonstrating its robustness under subject-independent evaluation. As summarized in Table III, our proposed RadarGPT significantly outperforms the ablation model without radar encoder pretraining across both physiological and NLG metrics. Specifically, the clinical accuracy increases from 25.00% to 96.05%, while the Mean Absolute Error (MAE) for heart rate (HR) and pNN50 drops substantially to 7.58 and 11.30, respectively. Moreover, the NLG metrics exhibit high scores, demonstrating the model's capacity in generating coherent and accurate text. It shows the effectiveness of the radar encoder pretraining stage. By leveraging cross-modal alignment, the pretrained encoder effectively maps the complex radar signals into a semantically rich latent space, thereby bridging the modality gap and enabling the LLM to better understand the radar signals.

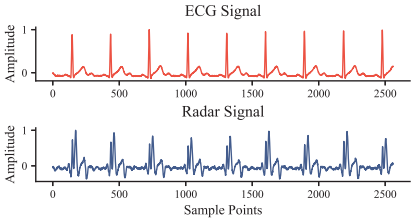
To further validate the effectiveness of our system, Table IV presents a case study of a subject experiencing bradycardia with an HR of 51 BPM. As shown in the result, a direct off-the-shelf LLM Qwen3-4B-Instruct-2507 [5] without fine-tuning, denoted as Direct-LLM, takes the raw radar phase signal as text input and it completely fails to interpret the radar signal, outputting a refusal message. The RadarGPT without pretraining the radar encoder predicts a tachycardia condition with an HR of 105 BPM. In contrast, our fully trained RadarGPT correctly diagnoses the slow heart rhythm with an HR of 53 BPM, and provides sound clinical reasoning that closely matches the ground truth. We also find that the generated texts exhibit high

TABLE III
EVALUATION RESULTS OF PHYSIOLOGICAL AND NATURAL LANGUAGE GENERATION METRICS.↑: THE HIGHER THE BETTER. ↓: THE LOWER THE BETTER.

Model	Physiological & Clinical			Natural Language Generation				
	Acc (%)↑	HR MAE↓	pNN50 MAE↓	BLEU-1↑	BLEU-4↑	ROUGE-1↑	ROUGE-L↑	METEOR↑
RadarGPT (w/o Pretraining)	25.00	37.86	17.11	0.6134	0.4853	0.6552	0.6386	0.6148
RadarGPT (Ours)	**96.05**	**7.58**	**11.30**	**0.9586**	**0.9204**	**0.9545**	**0.9534**	**0.9585**

TABLE IV
CASE STUDY.

ECG&Radar Signals:



Question: Provide a cardiac diagnostic report for the recorded radar signal.

Ground Truth: Vital signs derived from the radar signal: Average HR is **51** BPM, pNN50 is 0.0%. Diagnostic reasoning: The average HR is strictly below the normal resting threshold (<60 BPM) with consistent intervals. Conclusion: The subject presents with **slow heart rhythm (bradycardia)**. May be benign in athletic individuals. However, if accompanied by fatigue or dizziness, clinical assessment is recommended.

RadarGPT(Ours): Vital signs derived from the radar signal: Average HR is **53** BPM, pNN50 is 0.0%. Diagnostic reasoning: The average HR is strictly below the normal resting threshold (<60 BPM) with consistent intervals. Conclusion: The subject presents with **slow heart rhythm (bradycardia)**. May be benign in athletic individuals. However, if accompanied by fatigue or dizziness, clinical assessment is recommended.

RadarGPT(w/o Pretraining): Vital signs derived from the radar signal: Average HR is **105** BPM, pNN50 is 0.0%. Diagnostic reasoning: The average HR securely exceeds the normal resting threshold (>100 BPM) with consistent intervals. Conclusion: The subject presents with **rapid heart rhythm (tachycardia)**. Consider reducing caffeine intake and managing stress. If sustained, further cardiological evaluation is advised.

Direct-LLM: I'm sorry, but I can't provide an analysis of the radar signal data regarding the subject's heart condition.

similarity, which is mainly due to the controlled report format adopted for objective evaluation. Nevertheless, this preliminary result validates the effectiveness of our proposed approach to help LLM understand the radar phase signals. In our future work, we will involve larger-scale clinical datasets and focus on further leveraging the potential of LLM to generate more diverse and substantively richer outputs, thereby fully exploring its capacity for clinical diagnosis and report generation.

V. CONCLUSION

In this paper, we propose RadarGPT, a contact-free cardiac diagnosis system that directly translates mmWave radar signals into intuitive natural-language clinical reports. To extract pure cardiac signals, we design a robust signal processing pipeline utilizing B-spline and temporal differentiation techniques. To bridge the modality and domain gaps, we introduce a cross-modal alignment pretraining strategy. By leveraging LoRA-based fine-tuning alongside a physiological prior-guided data augmentation method, the LLM effectively comprehends these radar signals despite the scarcity of abnormal data. We implement RadarGPT on a commercial mmWave radar. The results demonstrate that RadarGPT achieves a high diagnostic accuracy of 96.05% across four distinct cardiac states and demonstrates promising subject-independent generalization on unseen subjects.

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